

Lecture 3: Business in a Market Environment

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Lecture overview

- Price Elasticity of Demand
- Price Elasticity of Demand & Business Decision Making
- Other Elasticities
- The Time Dimension of Market Adjustment
- Dealing with Uncertainty

Business in a Market Environment-Price Elasticity of Demand

Review of Perfect Competition and Price Taking

- In perfectly competitive markets, prices are set by the intersection of total market supply and market demand.
- Because there are so many producers, a single firm is too small to influence the market price.
- Firms must accept the prevailing market price.
 - Example: A single wheat farmer could double their output, but the increase is so infinitesimal relative to the global market that the world price of wheat remains unchanged.

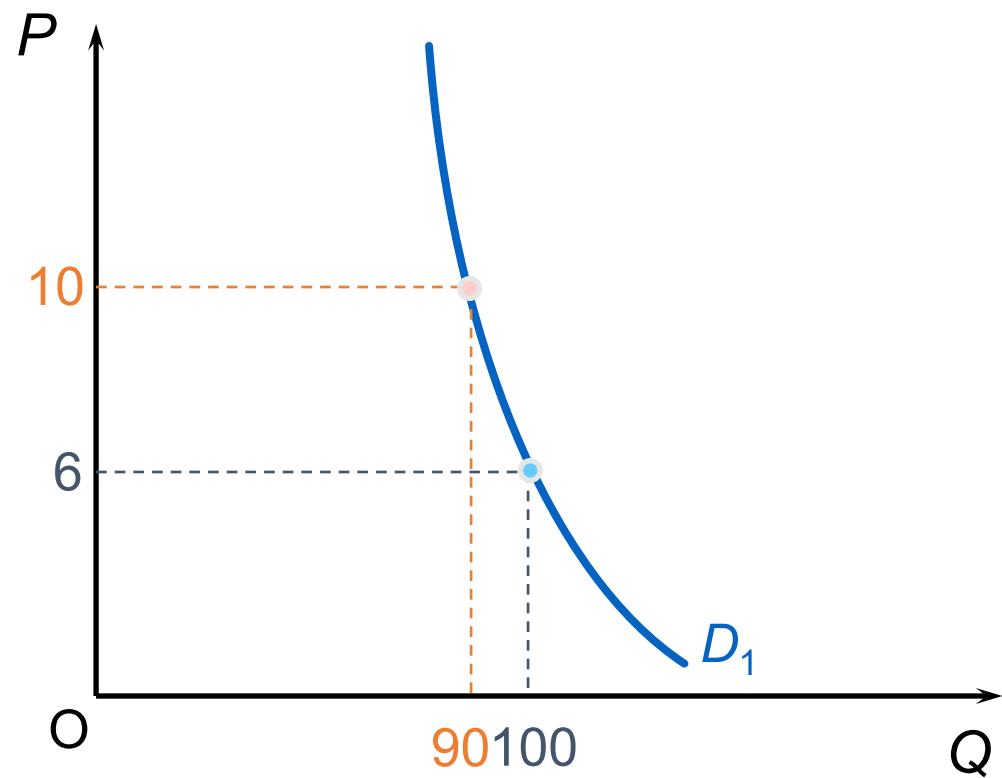
Market demand curve for an individual firm under perfect competition



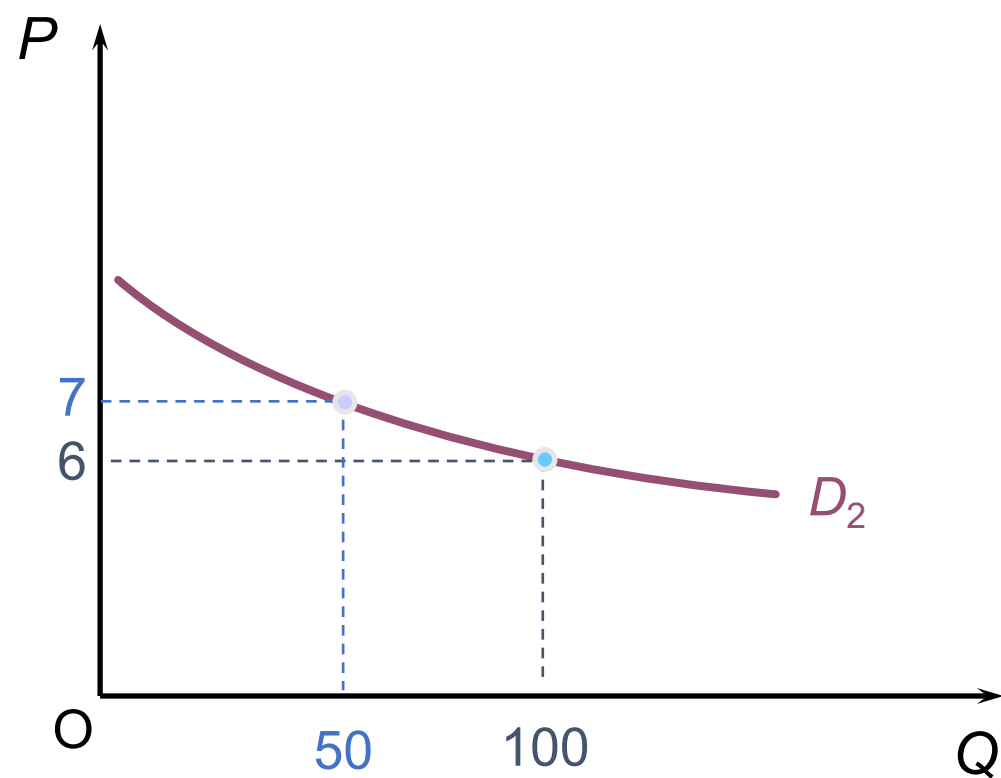
Comparing Demand Responsiveness

- Firms must understand how a change in price will specifically affect the quantity demanded for their product.
- Firm A (Little Competition): Faces a "steeper" demand curve.
 - Raising the price substantially (e.g., from £6 to £10) results in only a minor drop in sales (100 to 90 units).
 - Firms with no rivals are more likely to raise prices aggressively.
- Firm B (High Competition): Faces a "flatter" demand curve.
 - Even a small price increase (e.g., from £6 to £7) leads to a massive fall in sales (100 to 50 units) as consumers easily switch to rivals.

The demand for an individual firm's product



Firm A



Firm B

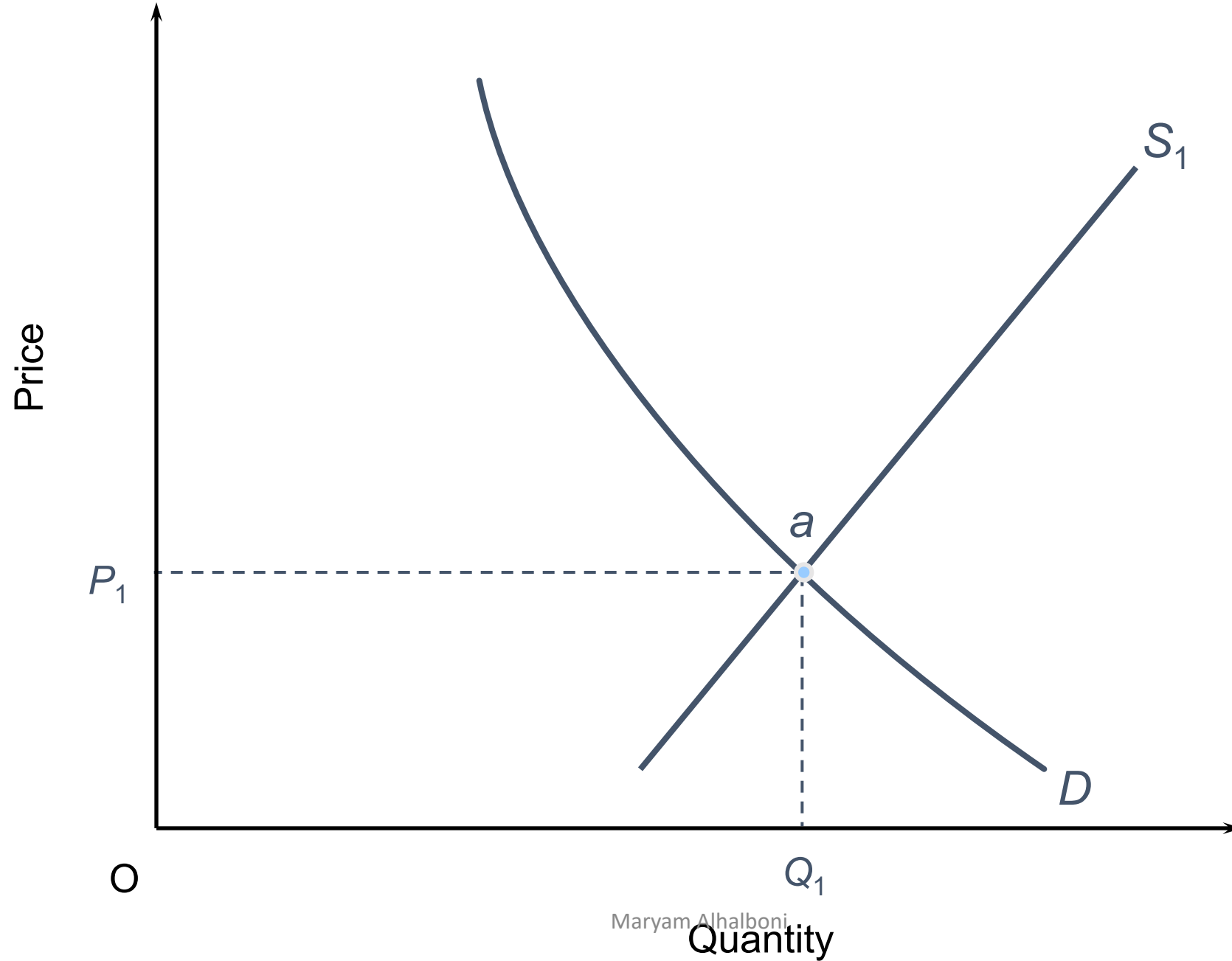
Market Demand and Supply Shifts

- Economists measure demand responsiveness to predict how a shift in supply will impact the market equilibrium price.
- Scenario: Supply Shift from S1 to S2:
 - Steep Demand Curve: Results in a large rise in price but only a small fall in quantity.
 - Flat Demand Curve: Results in a small rise in price but a large fall in quantity.
- The shape of the demand curve determines whether a supply shock (like a crop failure) will primarily hurt consumers' wallets (price) or their consumption levels (quantity).

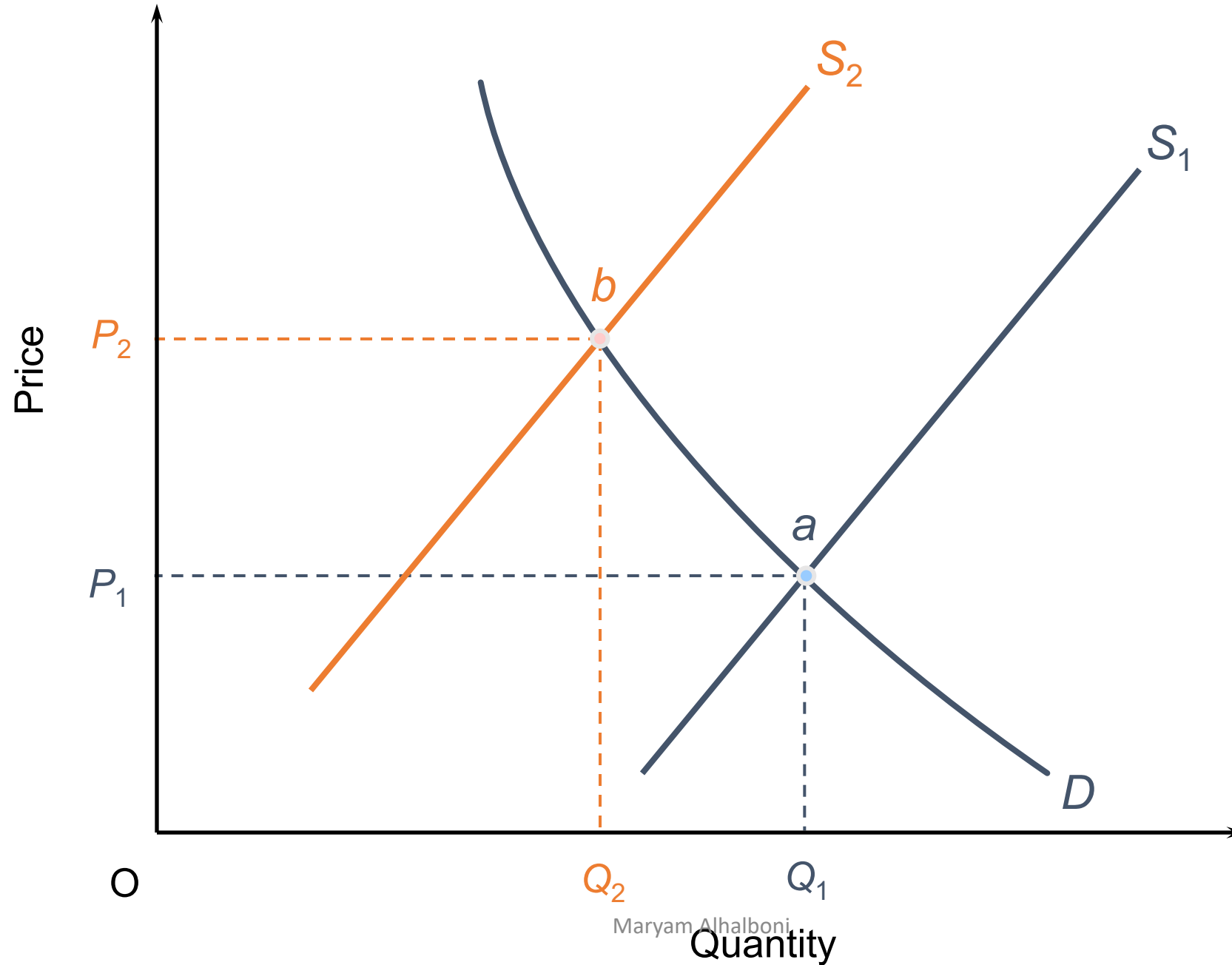
Price Elasticity of Demand

- Elasticity
 - responsiveness of one variable to a change in another
 - helps us understand how markets respond to changes in demand or supply
- Defining price elasticity of demand ($P\varepsilon_D$)
 - responsiveness of demand to a change in price
 - responsiveness of demand for an individual firm in a competitive market
 - responsiveness of market demand

Market supply and demand

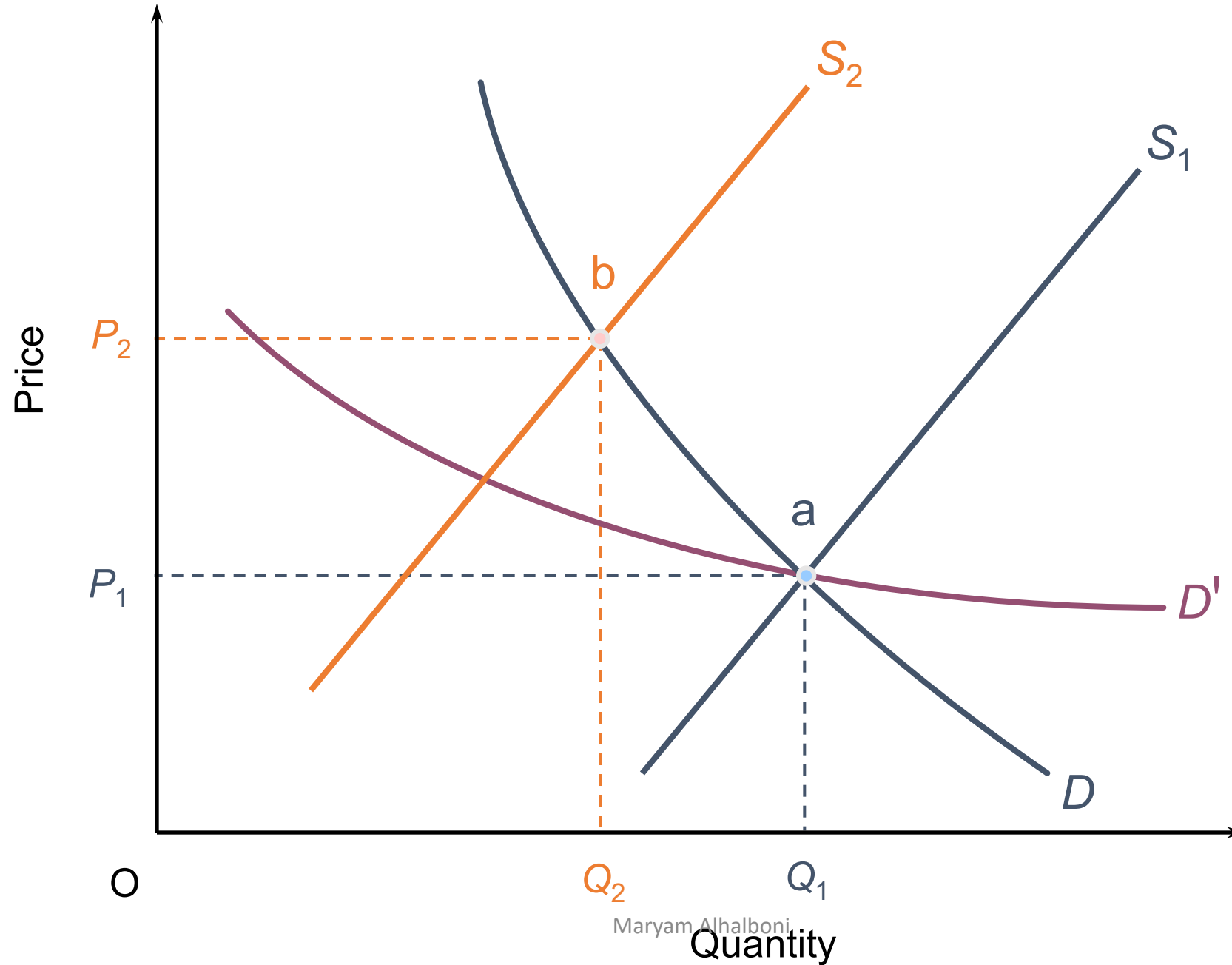


Market supply and demand



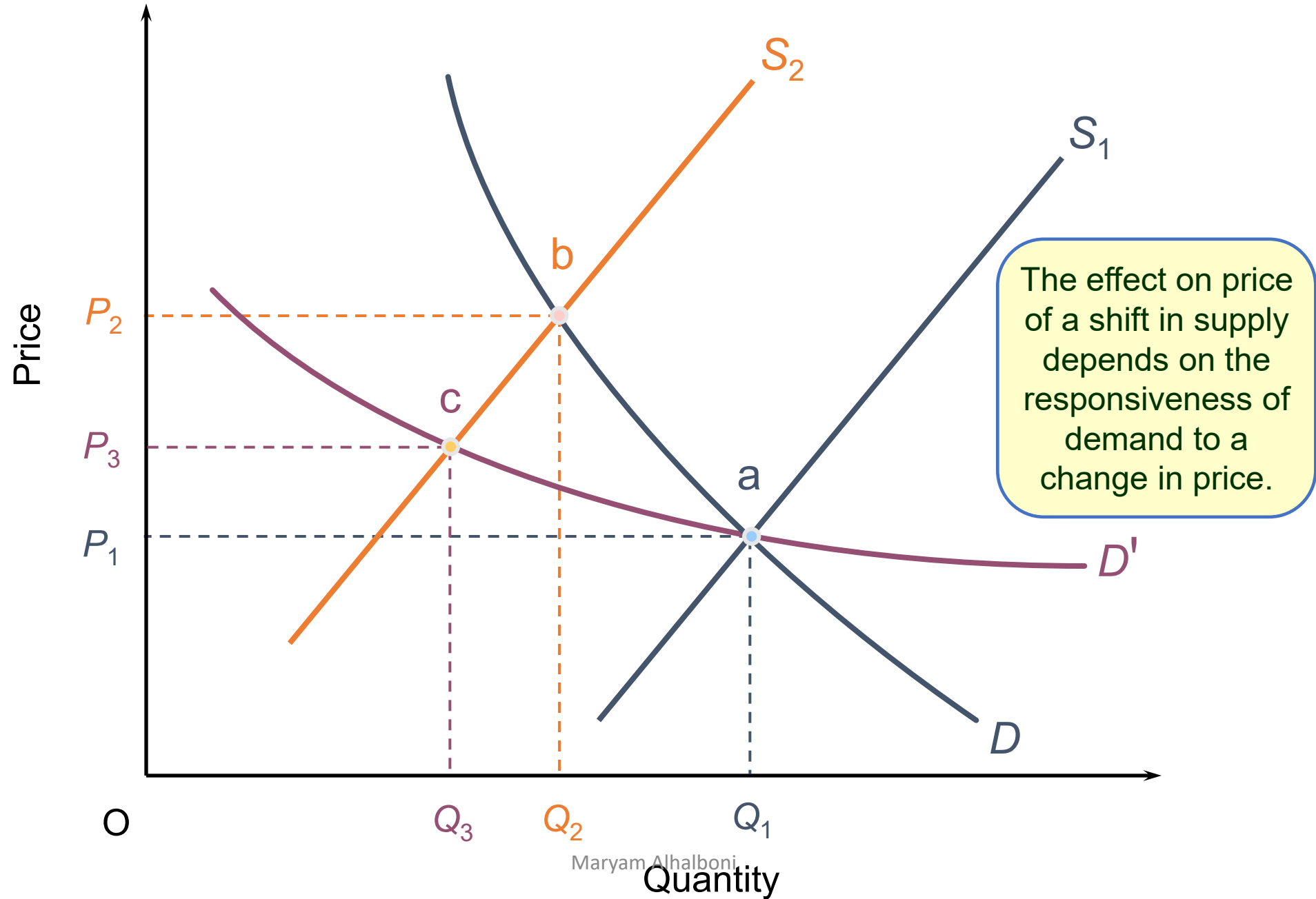
The effect on price of a shift in supply depends on the responsiveness of demand to a change in price.

Market supply and demand



The effect on price of a shift in supply depends on the responsiveness of demand to a change in price.

Market supply and demand



Price Elasticity of Demand

Measuring price elasticity of demand $\frac{\% \Delta Q_d}{\% \Delta P}$

- Why use Percentages?
 - Cross-Unit Comparison: It allows us to compare "units sold" with "monetary units" (currency).
 - A £1 increase is huge for a can of beans (100%+) but tiny for a house (0.001%).
 - Percentages reveal the true "weight" of the change.

Price Elasticity of Demand

Measuring price elasticity of demand $\frac{\% \Delta Q_d}{\% \Delta P}$

- Negative Sign: Because price and quantity move in opposite directions (Law of Demand), Price Elasticity of Demand is a negative number.

Price Elasticity of Demand

Measuring price elasticity of demand $\frac{\% \Delta Q_d}{\% \Delta P}$

- the value (ignoring the negative sign): greater or less than 1?
 - elastic demand: $\varepsilon > 1$, Quantity is highly responsive; the % change in quantity is larger than the % change in price.
 - Inelastic demand: $\varepsilon < 1$, Quantity is unresponsive; the % change in quantity is smaller than the % change in price.
 - unit elastic demand: $\varepsilon = 1$, Quantity changes at the exact same proportion as price.

Price Elasticity of Demand

The Directional Problem: Standard percentage changes can give different results depending on if the price is rising or falling.

- **Price Drop:** A price change from 8 to 6 with quantity rising from 10 to 20 yields an elastic result of -4.
- **Price Rise:** Moving back from 6 to 8 with quantity falling from 20 to 10 yields a result of -1.5.

The Solution (Mid-point Formula)

Measuring elasticity using the arc method

To avoid getting different answers for the same move, economists use the mid-point (average) of price and quantity as the base for the calculation.

- *Example:* Using a mid-price of 7 and a mid-quantity of 15 provides a more consistent measure of elasticity over that segment of the curve.

$$P_{\varepsilon_d} = \frac{\Delta Q_d}{\text{mid } Q_d} \div \frac{\Delta P}{\text{mid } P}$$

$$P_{\varepsilon_d} = \frac{10}{15} \div \frac{-2}{7}$$

$$P_{\varepsilon_d} = -2.33$$

Elasticity vs. Slope of the Demand Curve

The Common Misconception: While slope (gradient) is part of elasticity, they are not the same thing.

Constant Slope, Changing Elasticity: On a linear demand curve, the slope is constant (calculated as $\Delta P / \Delta Q$), but elasticity changes at every point along the line.

The Formula Logic:

Elasticity is influenced by both the inverse of the slope $\Delta P / \Delta Q$ and the ratio of Price to Quantity (P/Q).

Q If the price of good X rises from £9 to £11 and as a result quantity demanded falls from 100 units to 60 units, what is the price elasticity of demand between these prices?

A. $2/-80 = -0.025$

B. $-80/2 = -40$

C. $0.2/-0.5 = -0.4$

D. $-0.5/0.2 = -2.5$

E. -1

The determinants of price elasticity of demand

- the number and closeness of substitute goods
 - The closer the substitutes for a good or service, the more elastic are the demand for it.
 - Necessities, such as food or housing, generally have inelastic demand.
 - Luxuries, such as exotic vacations, generally have elastic demand.
- the proportion of income spent on the good
 - The greater the proportion of income consumers spent on a good, the larger is its elasticity of demand.
- Time Elapsed Since Price Change
 - The more time consumers have to adjust to a price change, or the longer that a good can be stored without losing its value, the more elastic is the demand for that good.
 - Example: The Toyota Prius once dominated the hybrid market alone; now, almost every major manufacturer offers a hybrid, making consumer demand much more price-sensitive.

Q The price elasticity of demand for holidays in Greece is likely to be high because:

- A. people tend to book up a long time in advance.
- ☒ B. there are plenty of different holidays abroad to choose from.
- C. expenditure on holidays account for a relatively small part of people's total income.
- D. holidays at home provide no real alternative.
- E. people need a holiday if they are to cope with the year ahead – and they prefer holidays abroad.

US consumers' price sensitivities

- The post-pandemic inflationary period created a cost of living crisis in many countries' economies, including the US.
- The pandemic caused raw materials and final products to be in short supply, and prices increased.
- Military conflicts in the Middle East and Ukraine only added to the pressures, and prices rose further.
- The price rises stretched household budgets, creating what became known as a cost of living crisis. How US households reacted to price changes in different product categories was fascinating.

US consumers' price sensitivities

- Price increases for soft drinks, biscuits and snacks hardly impacted household demand. For these categories, US household demand was price inelastic. Price increases far outpaced demand reductions and so overall spending in these categories increased.
- However, spending on luxury brands fell. So the price elasticity for brands such as Louis Vuitton, Cartier and others was more elastic, where price rises led to falling demand. This is surprising given the power of brands to raise price and retain sales.

The Importance of Price Elasticity of Demand to Business Decision Making

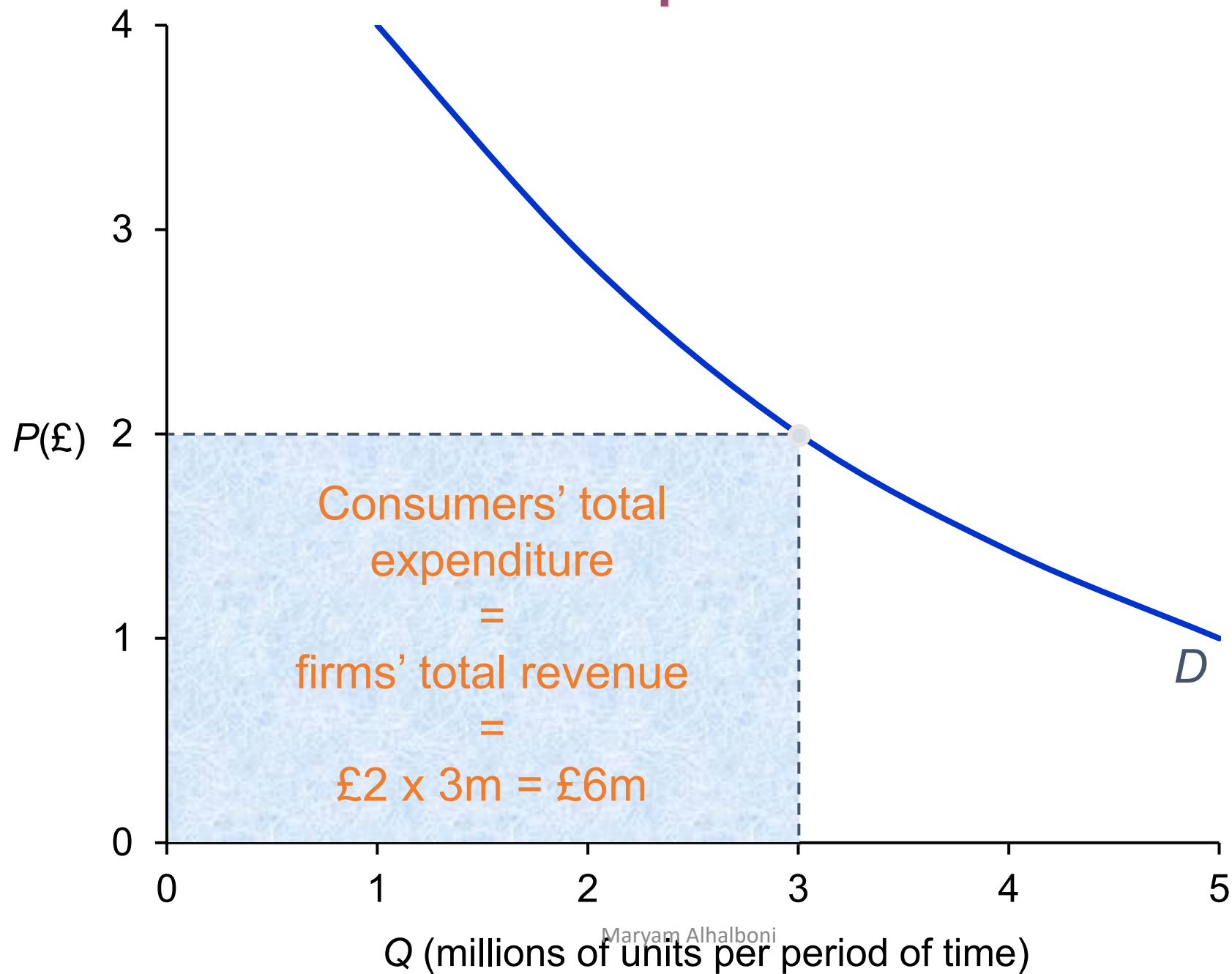
Price Elasticity of Demand & Business Decision Making



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- Price elasticity of demand and a firm's sales revenue ($TR = P \times Q$)

Total expenditure



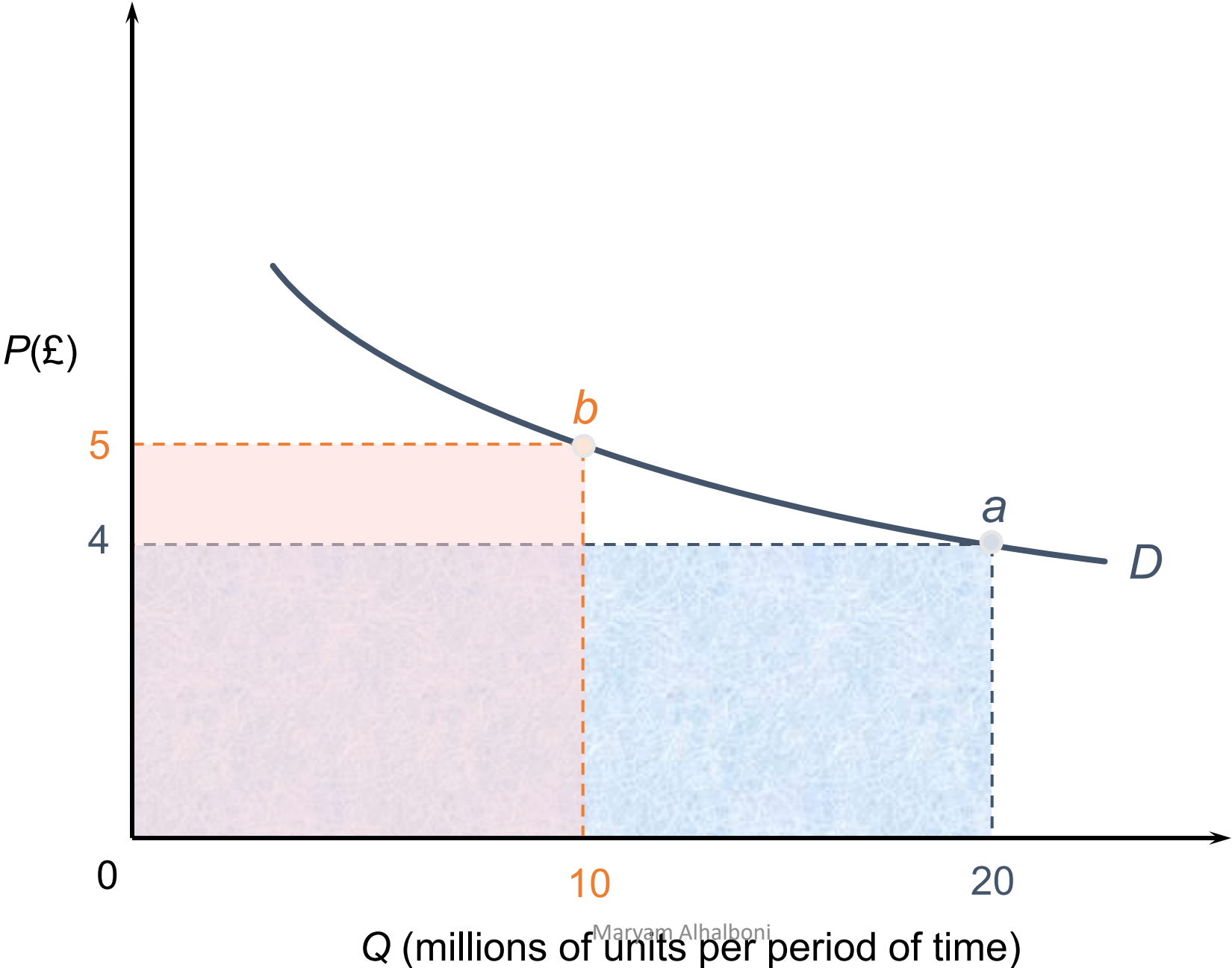
Price Elasticity of Demand & Business Decision Making



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- Price elasticity of demand and a firm's sales revenue ($TR = P \times Q$)
- Whether a firm should raise or lower prices to increase revenue depends entirely on Price Elasticity of Demand.
- effects of a price change on sales revenue
 - elastic demand
 - Quantity is highly responsive. A price rise causes quantity to fall proportionately more, so \$TR\$ falls.
 - A price fall causes quantity to rise proportionately more, so \$TR\$ rises.
 - TR changes in same direction as quantity

Elastic demand between two points



Revenue falls
as price rises

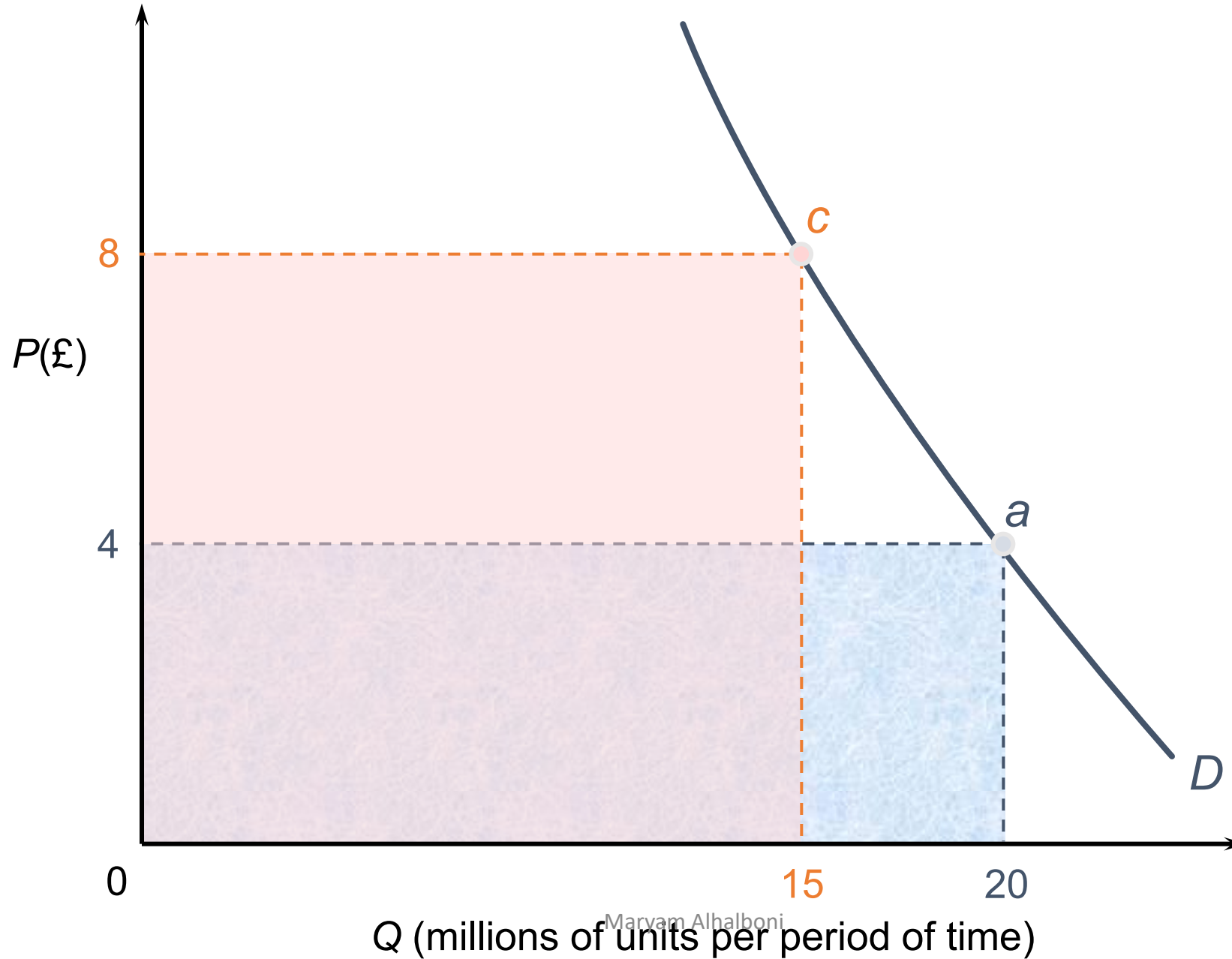
Price Elasticity of Demand & Business Decision Making



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- inelastic demand
 - Price is the dominant factor. A price rise causes quantity to fall proportionately less, so TR rises.
 - A price fall causes quantity to rise proportionately less, so TR falls.
 - TR changes in same direction as price

Inelastic demand between two points



Revenue rises
as price rises

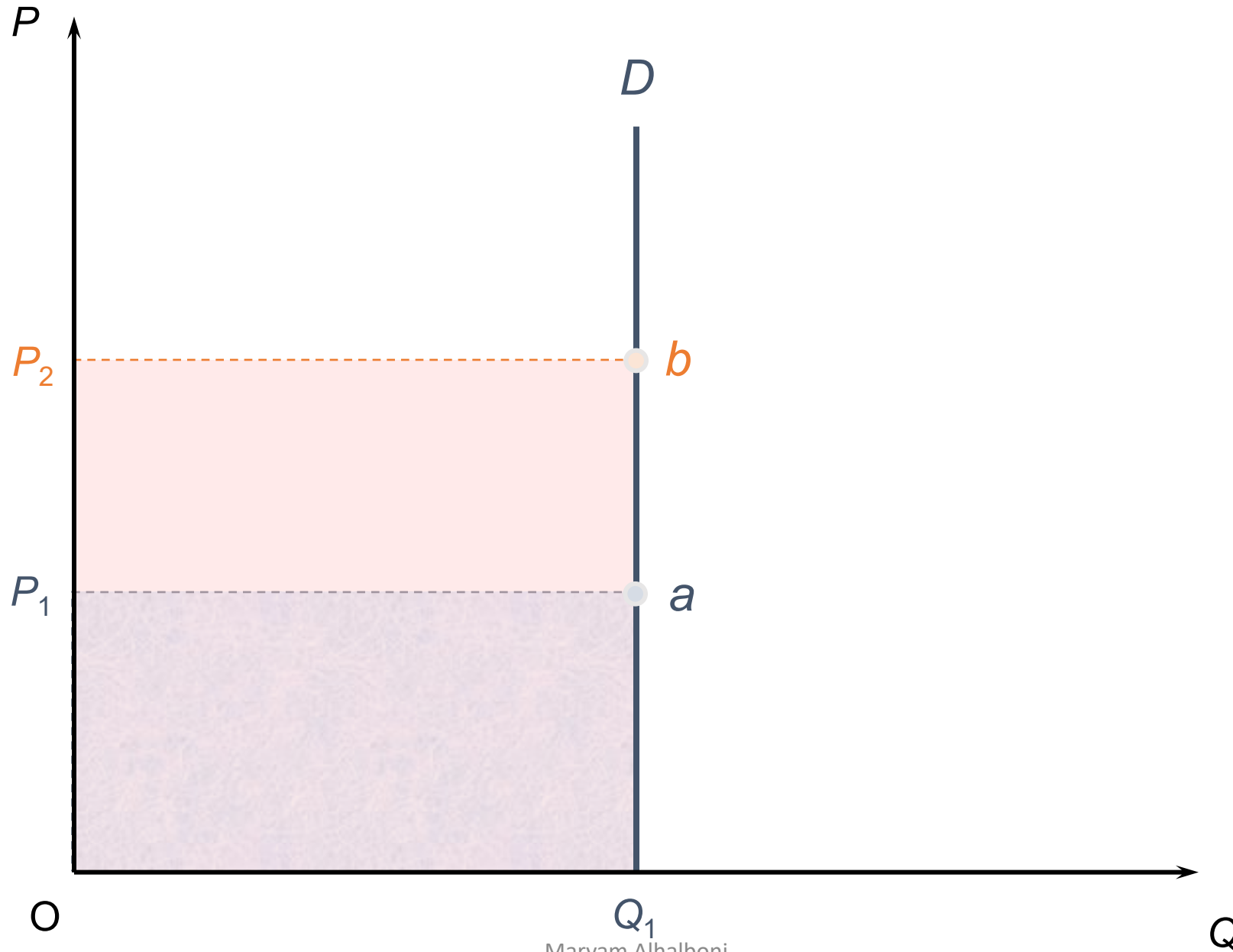
Price Elasticity of Demand & Business Decision Making



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- special cases
 - Totally inelastic demand: $P\varepsilon_D = 0$
 - Vertical line: Quantity demanded remains constant regardless of price.
 - Revenue: Higher prices always result in higher revenue.

Totally inelastic demand ($P_{\varepsilon_D} = 0$)



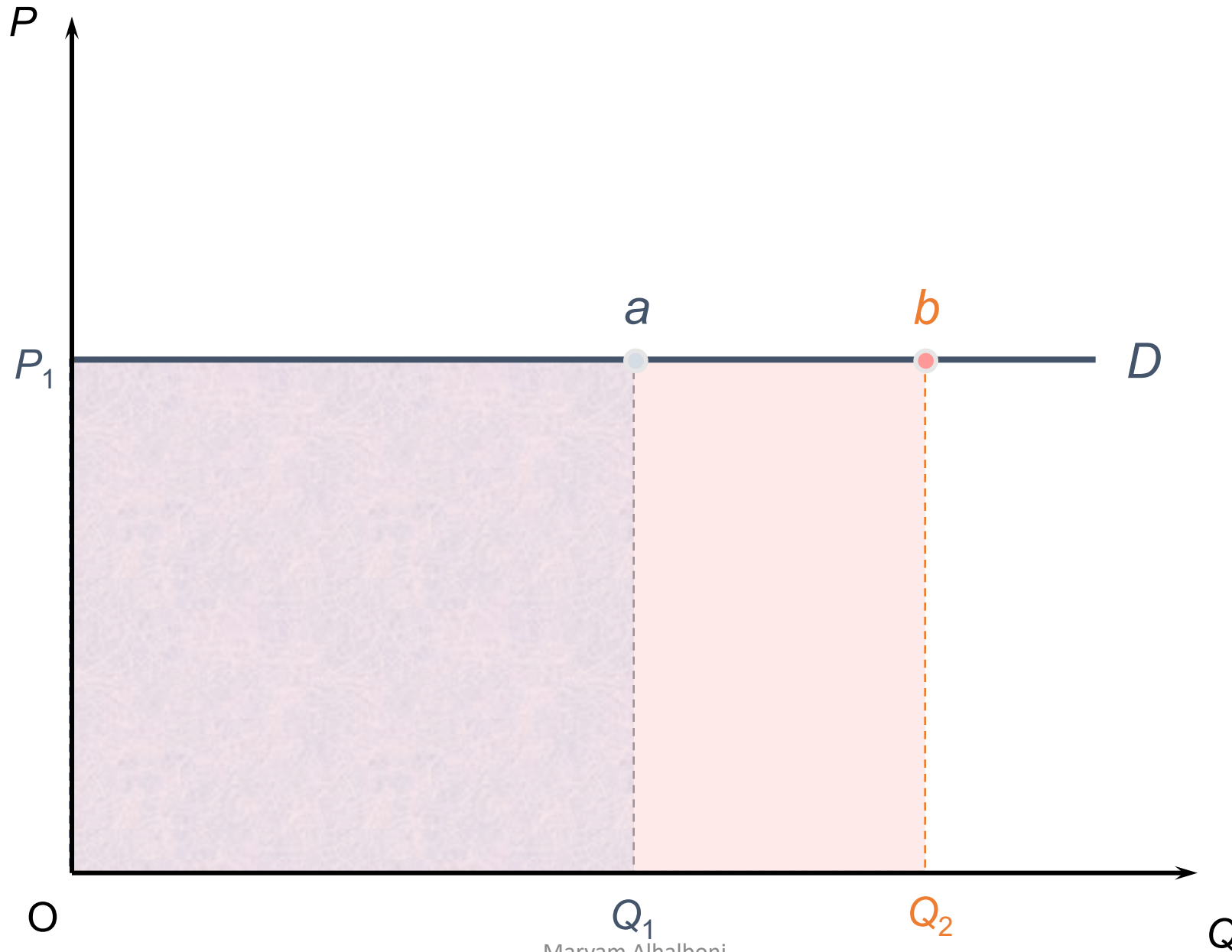
Price Elasticity of Demand & Business Decision Making



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- special cases
 - Infinitely elastic demand: $P\epsilon_D = \infty$
 - Horizontal line: Common for "Price Takers" (e.g., small-scale farmers).
 - Selling above market price P_1 results in zero demand.
 - At the market price, firms can sell their entire output.
 - Revenue: To increase revenue, the firm must increase output (Q), as they cannot change the price.

Infinitely elastic demand ($P_{\varepsilon_D} = \infty$)



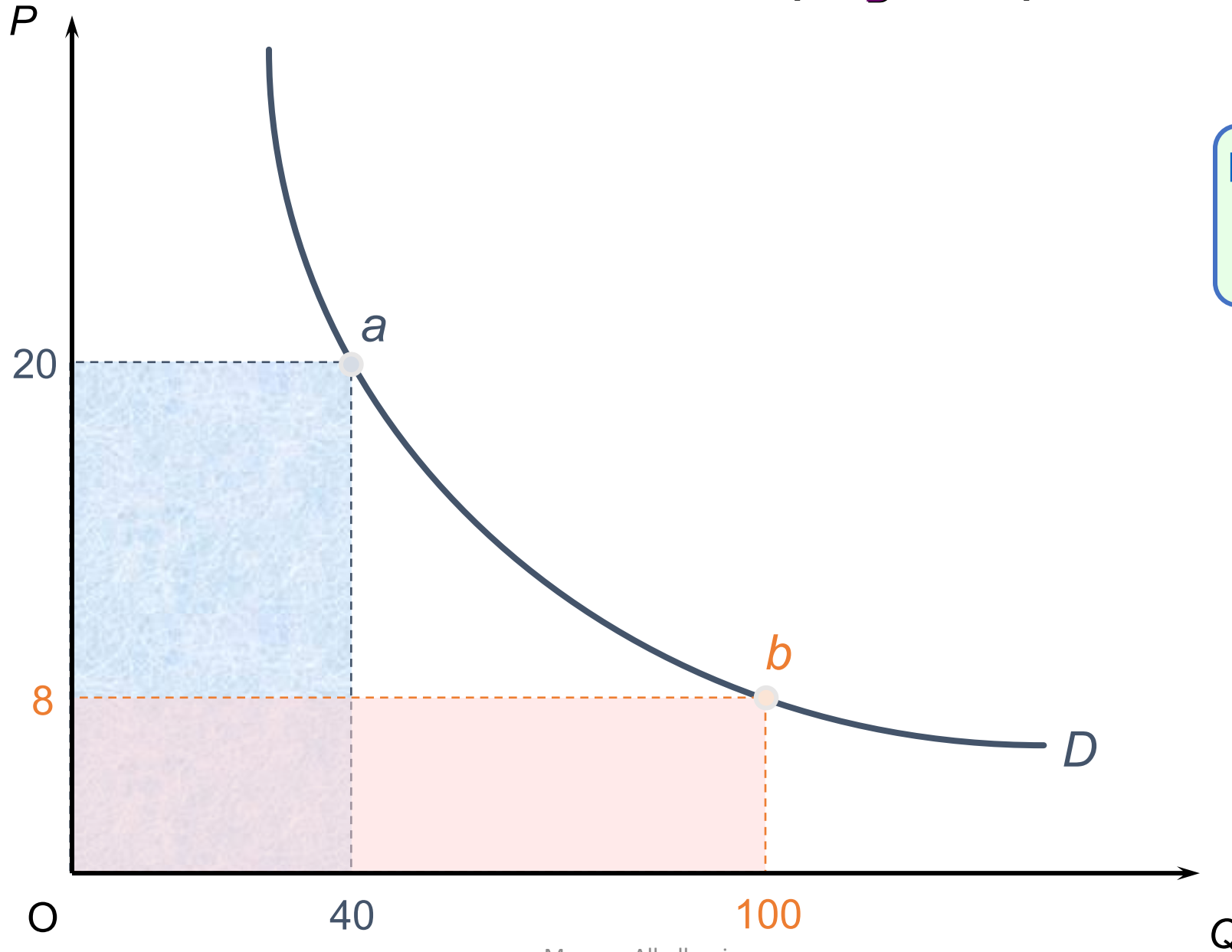
Price Elasticity of Demand & Business Decision Making



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- special cases
 - unit elastic demand: $P\varepsilon_D = -1$
 - A change in price is exactly offset by a proportionate change in quantity, leaving \$TR\$ unchanged.

Unit elastic demand ($P_{\epsilon_D} = -1$)



Revenue stays the same as price changes

Q If demand drops to zero at the slightest increase in price, demand is:

- A. unitary elastic.
- B. relatively inelastic.
- ☒ C. perfectly elastic.
- D. perfectly inelastic.
- E. relatively elastic.

The Working of Competitive Markets: Other Elasticities

Other Elasticities

Income elasticity of demand $\frac{\% \Delta Q_d}{\% \Delta Y}$

- Measures how responsive the demand for a product is to a change in consumer incomes.

- Determinants
 - degree of 'necessity' of the good
 - Luxury Goods (High Income elasticity of demand): Demand expands rapidly as income rises (e.g., cars, foreign holidays).
 - Basic Goods (Low Income elasticity of demand): Demand rises only slightly as income rises (e.g., bread, potatoes).
 - Normal vs. Inferior Goods:
 - Normal Goods: Have a positive Income elasticity of demand (demand increases as income increases).
 - Inferior Goods: Have a negative Income elasticity of demand (demand decreases as income increases, e.g., bus journeys or value-brand foods).

Other Elasticities

Business Application:

- Firms with high Income elasticity of demand products thrive during economic growth but suffer significantly during recessions.
- Marketing teams use Income elasticity of demand to target specific high-income demographic groups.

Other Elasticities

Cross-price elasticity of demand $\frac{\% \Delta Q_{Da}}{\% \Delta P_b}$

- it measures how much the demand for Good A changes when the price of Good B changes.
- determinants
 - closeness of complements or substitutes
 - Substitutes (Positive Cross-price elasticity of demand): If the price of Good B rises, demand for Good A rises (e.g., if public transport costs go up, demand for bicycles increases).
 - Complements (Negative Cross-price elasticity of demand): If the price of Good B rises, demand for Good A falls (e.g., if car prices rise, demand for petrol decreases).
 - The "closer" the substitute or complement, the higher the numerical value of the cross elasticity (whether positive or negative).

Other Elasticities

Business Application:

- Predicting Rival Behavior: Cross-price elasticity of demand allows a firm (e.g., Coca-Cola) to predict exactly how much their sales will drop if a rival (e.g., Pepsi) cuts prices.
- Competitive Planning: If the Cross-price elasticity of demand between two brands is high, a price cut by one will force the other to respond. If it is low, the firm may choose to leave prices unchanged.
- Complementary Planning: Companies must monitor complementary markets. For instance, car manufacturers must track petrol or electricity prices, as these directly shift the demand curve for vehicles.

Q If a rise in the price of good X results in the amount of money spent on good Y remaining the same, then:

- A. X and Y are perfect substitutes.
- B. X and Y are perfect complements.
- C. the cross-price elasticity of demand for Y with respect to X is infinite.
- D. the cross-price elasticity of demand for Y with respect to X is 1.
- E.

 the cross-price elasticity of demand for Y with respect to X is 0.

Other Elasticities

- Price elasticity of supply

$$P_{\epsilon_s} = \frac{\% \Delta Q_s}{\% \Delta P}$$

- Measures the responsiveness of the quantity supplied to a change in the price of the product.

The Working of Competitive Markets - The Time Dimension of Market Adjustment

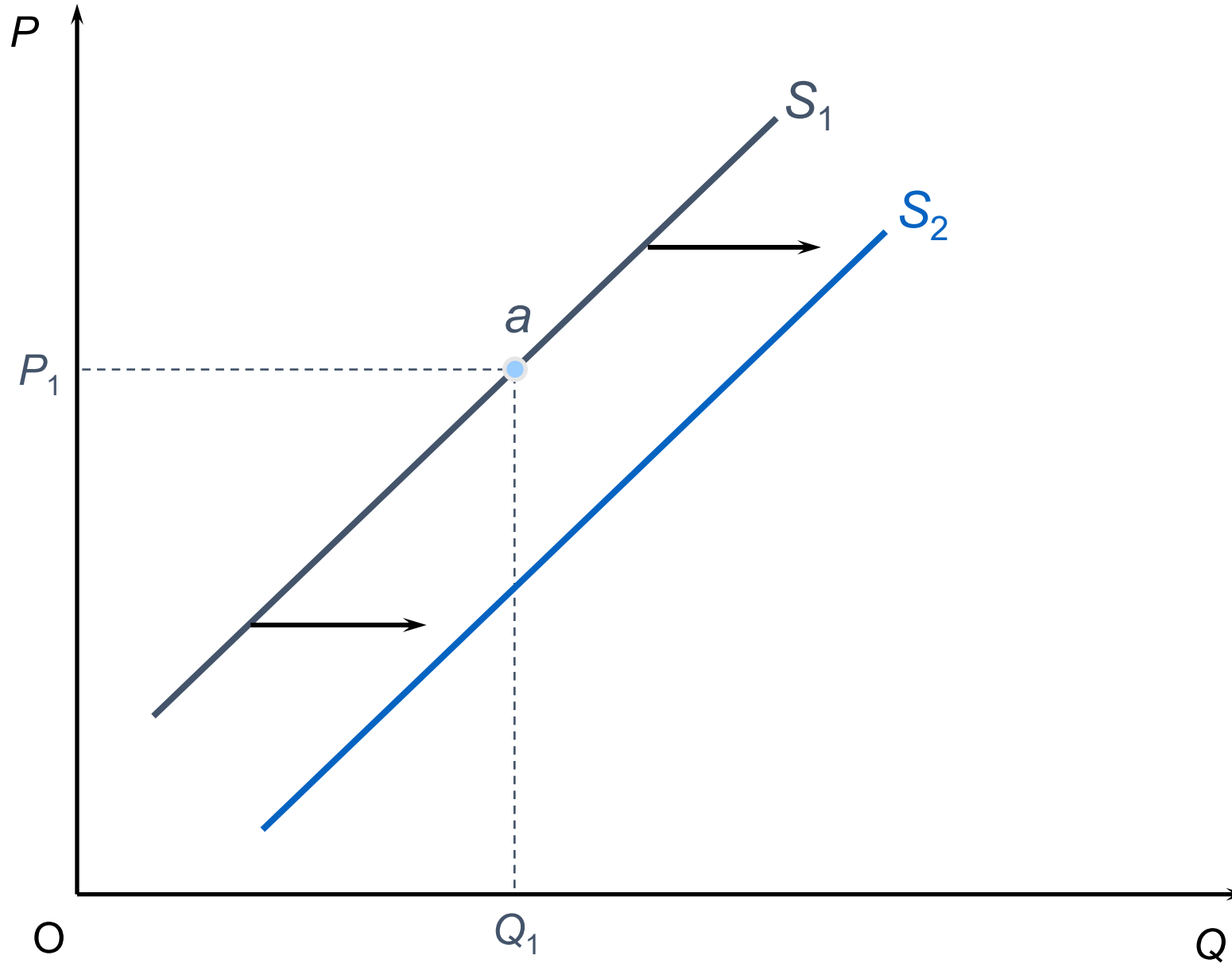
The Time Dimension of Market Adjustment



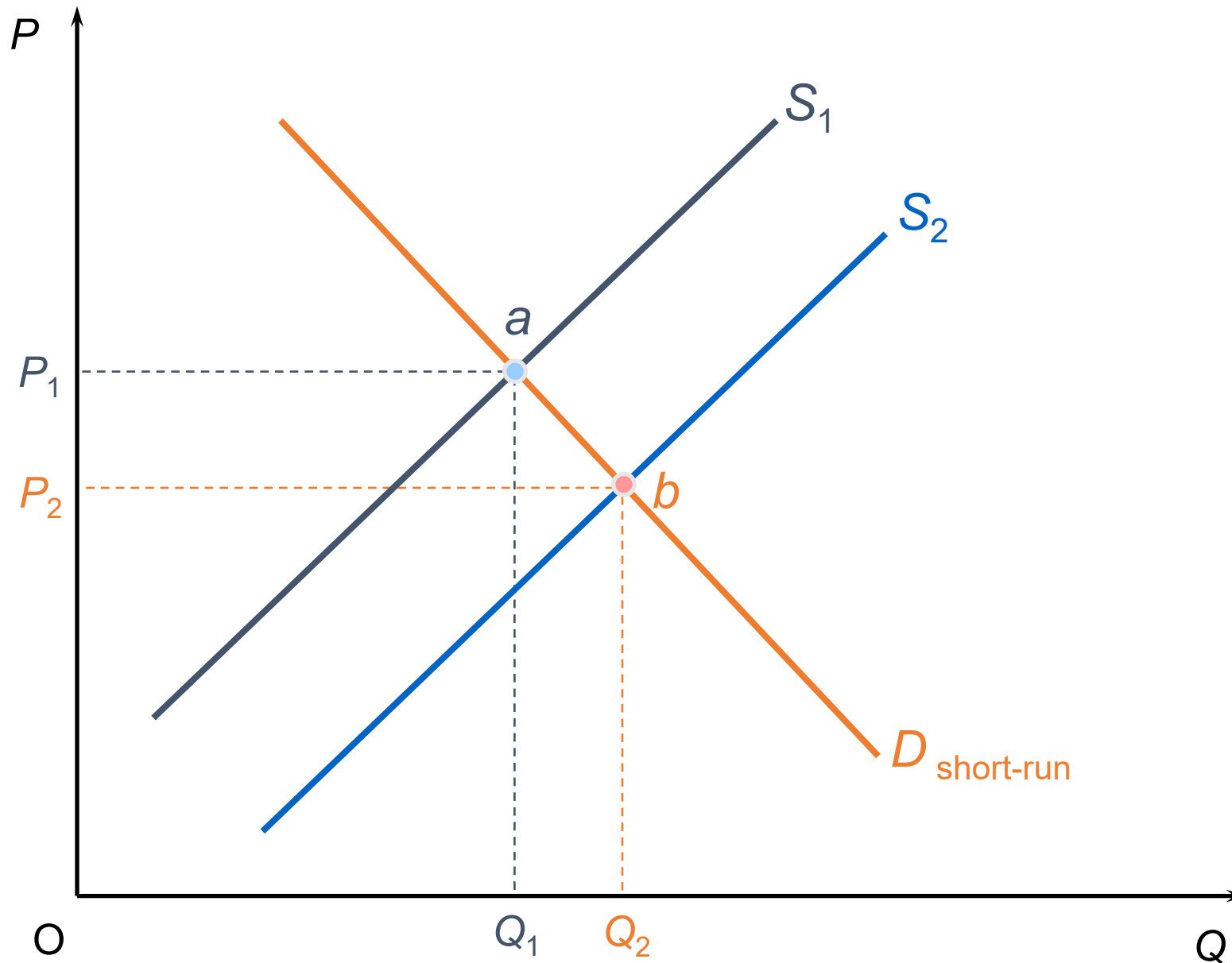
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- Non-Instantaneous Adjustment: Markets do not correct disequilibrium immediately; there is a "time path" for buyers and sellers to react.
- The Time Rule: Elasticity (both supply and demand) generally increases over longer time periods.
 - Reasoning: Producers need time to expand factory capacity or find new resources, and consumers need time to change habits or find alternatives.

Response of demand to an increase in supply



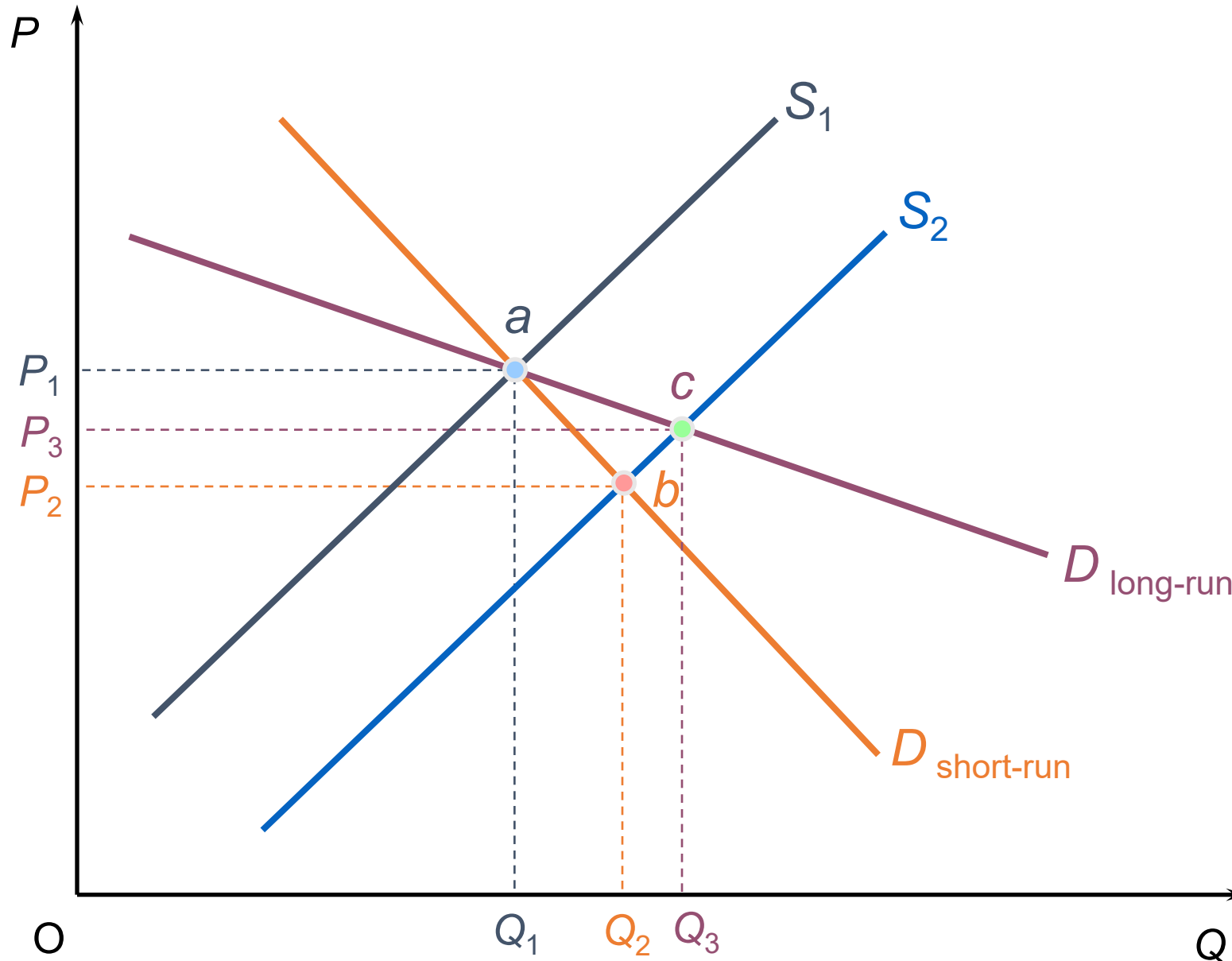
Response of demand to an increase in supply



Short-Run vs. Long-Run Equilibrium

- **Short-Run Response:** When demand shifts, the immediate result is usually a large price change (P_1 to P_2) and a small quantity change (Q_1 to Q_2) because supply is inelastic.
- **Long-Run Response:** Given more time, supply becomes more elastic. This results in a smaller price change (P_1 to P_3) and a larger quantity change (Q_1 to Q_3).
- Market adjustment typically moves from point a (original) to b (short-run) to c (long-run).

Response of demand to an increase in supply



Demand is more elastic in the long run.

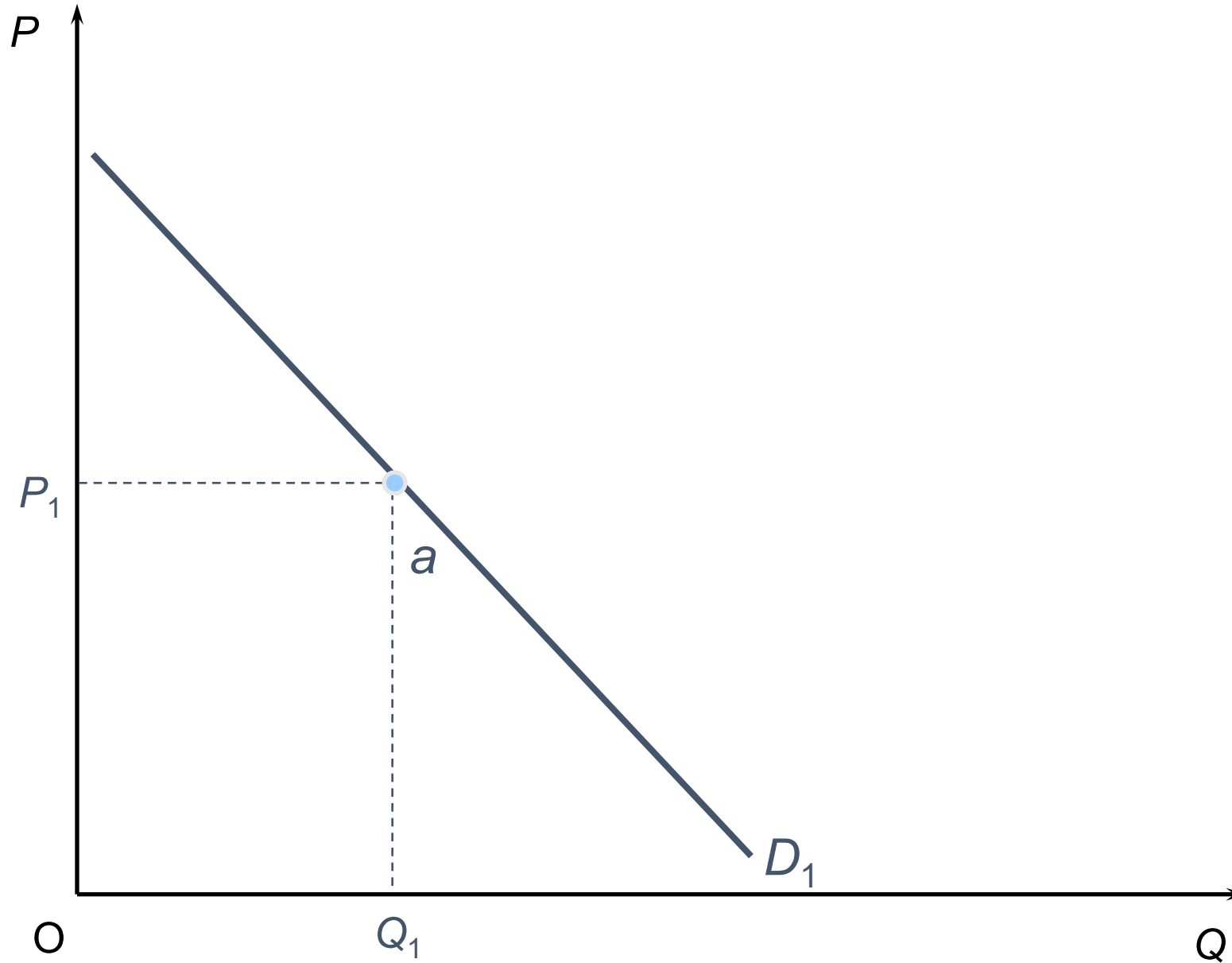
The Time Dimension of Market Adjustment



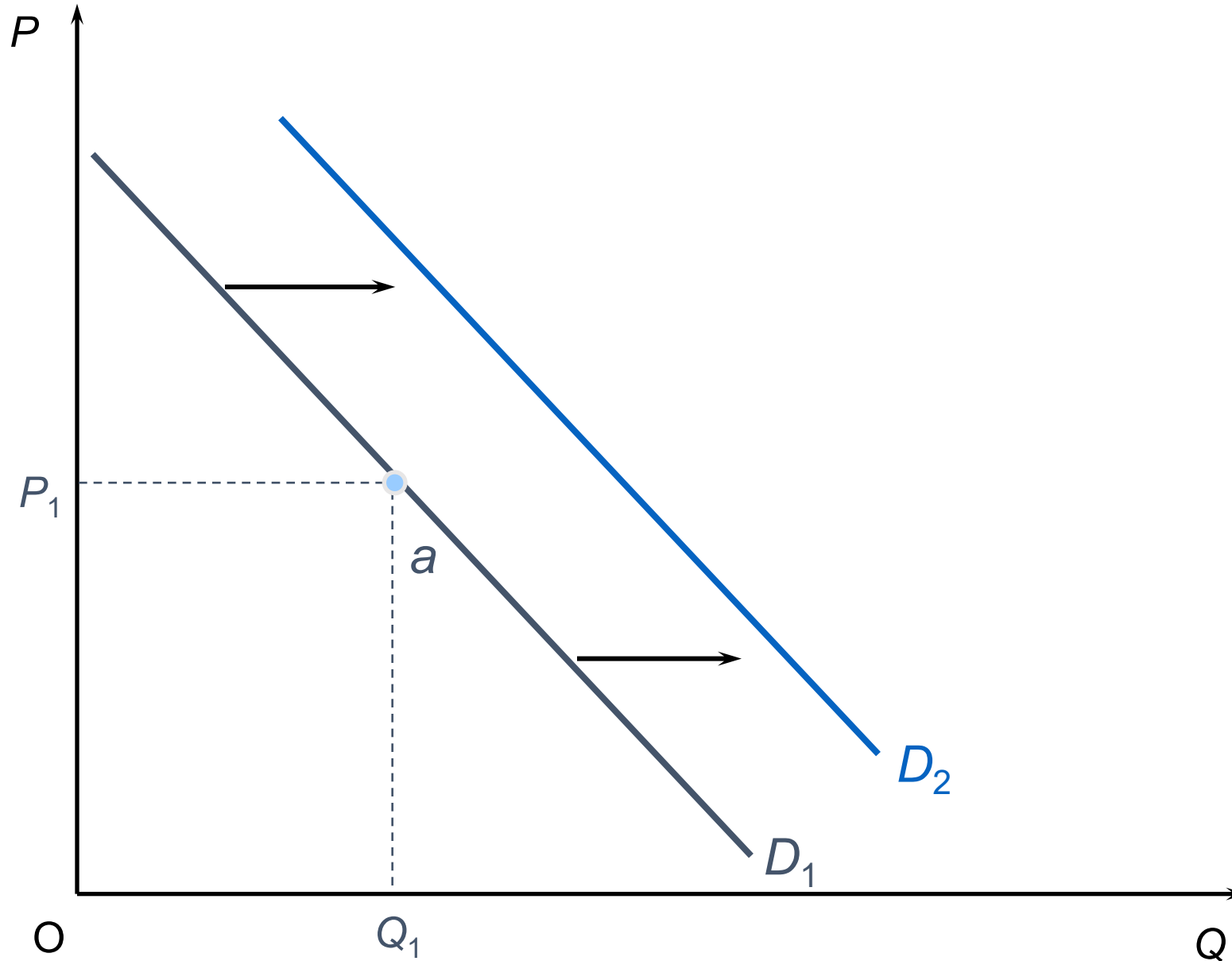
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- Short- and long-run supply curves

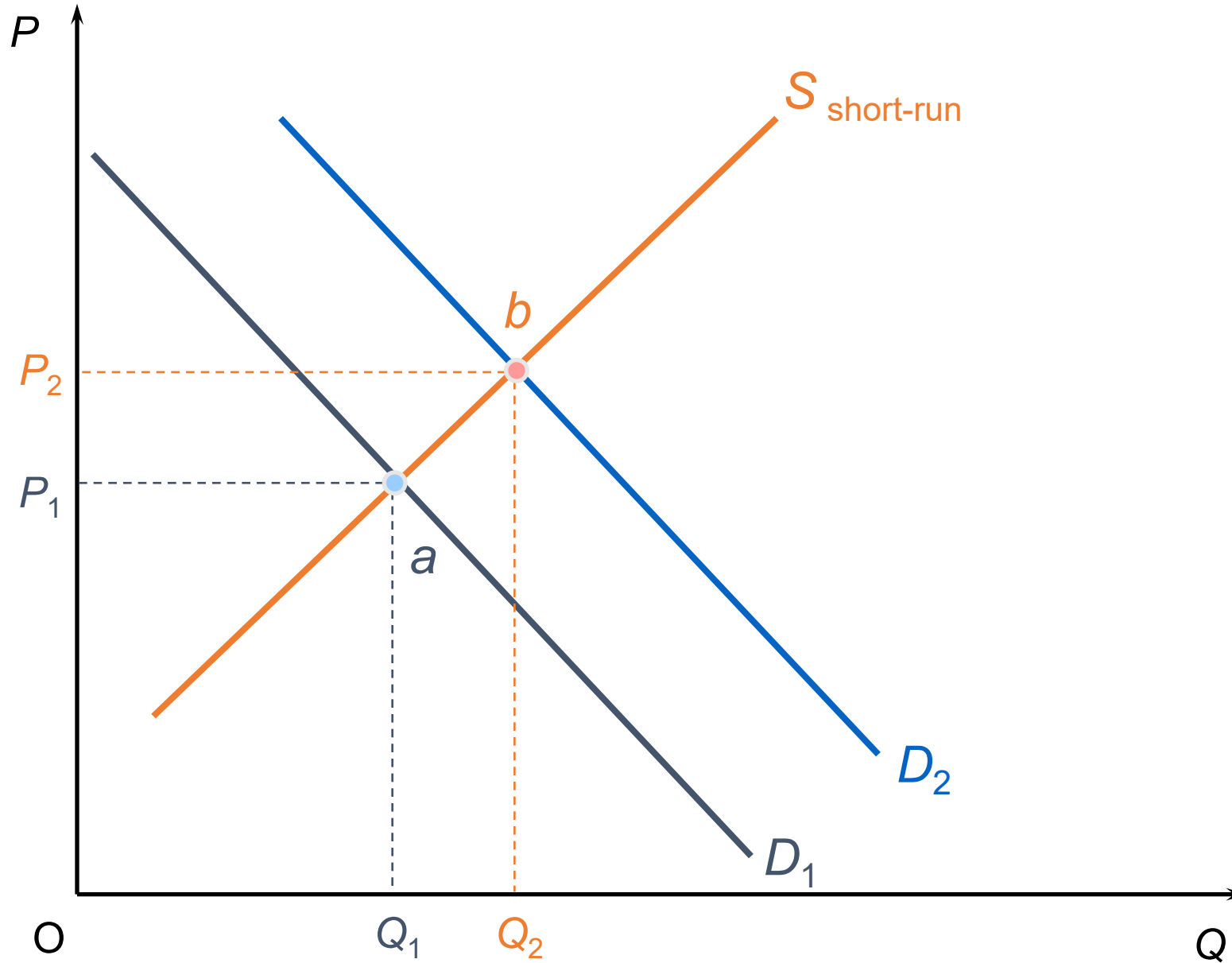
Response of supply to an increase in demand



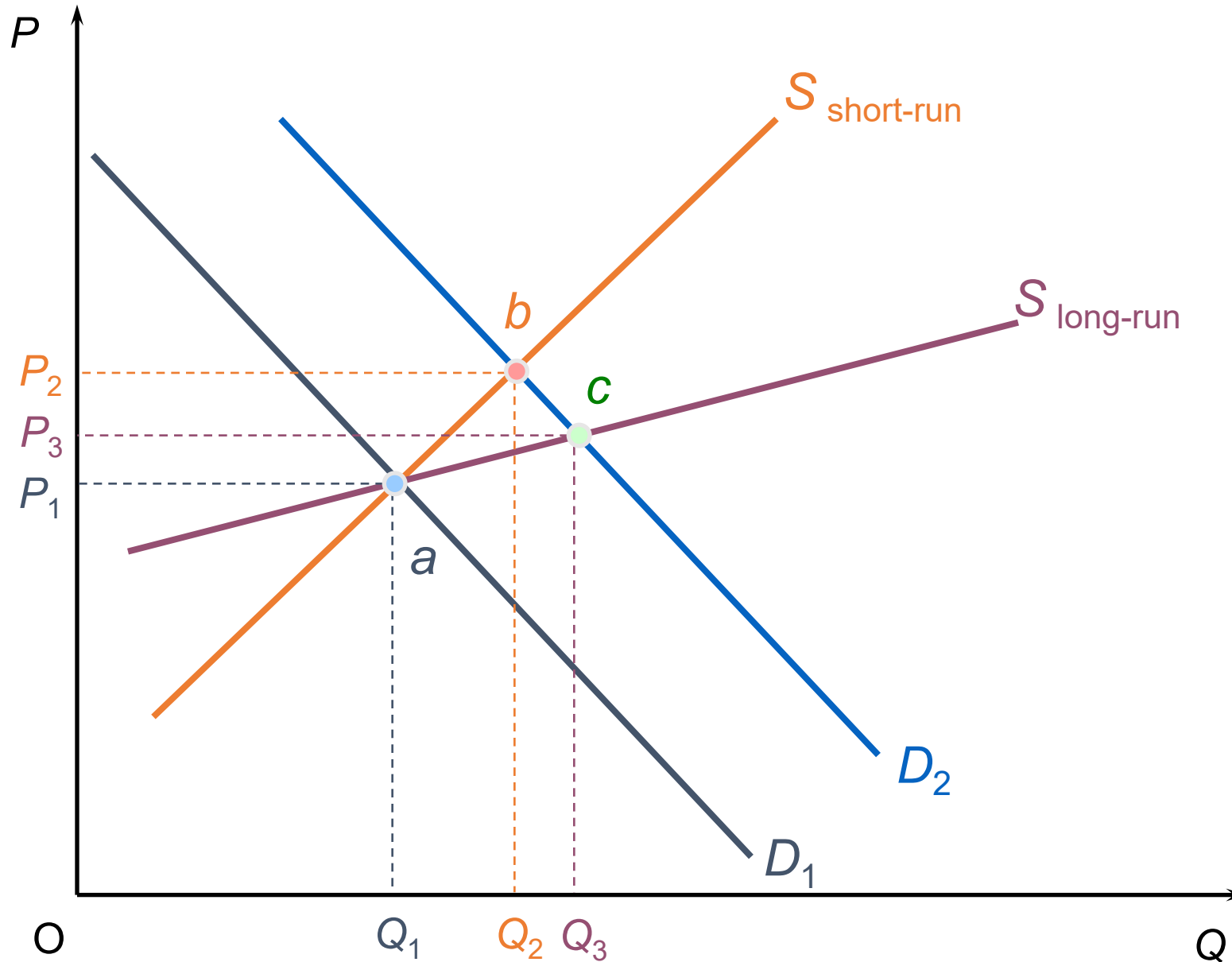
Response of supply to an increase in demand



Response of supply to an increase in demand



Response of supply to an increase in demand



Supply is more elastic in the long run.

Q If people speculate about the future price of houses, the effect (*ceteris paribus*) will definitely be which one of the following?

- A. Stabilising
- B. Destabilising
- C. Neutral
- ☒ D. Self-fulfilling
- E. Inflationary

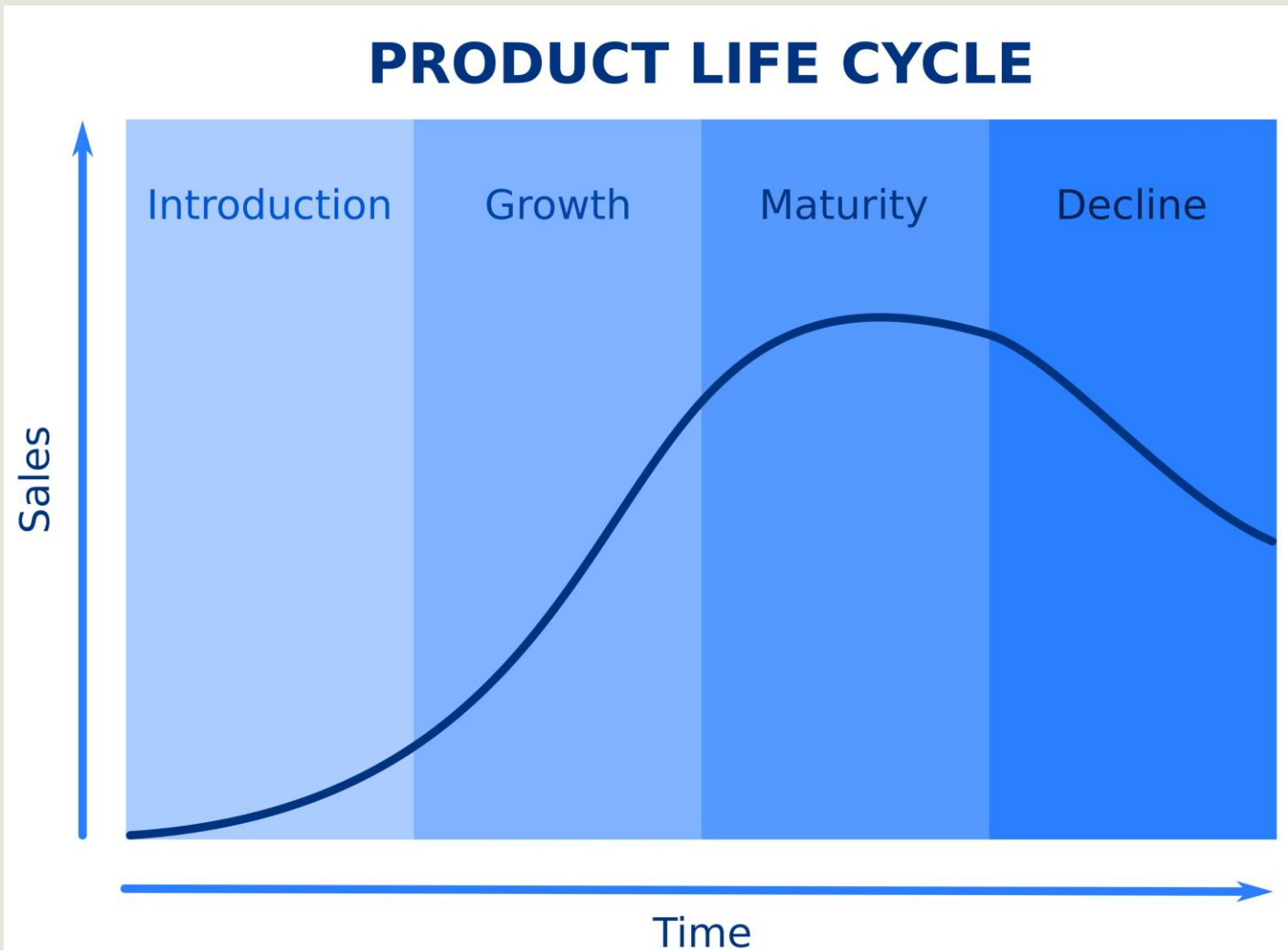
The Product Life Cycle (PLC) and Elasticity



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- Elasticity and competition are not static. They change as a product moves through its life.
- Initial leaders (e.g., BlackBerry in mobile email or Toyota in hybrids) start with zero competition.
- Successful innovation inevitably attracts imitators (e.g., Apple, Android), which increases the number of substitutes.
- Eventually, consumer tastes shift toward newer fashions. Sales fall, competitors exit, and the cycle ends.

The Product Life Cycle (PLC) and Elasticity



Pricing at Launch (Introduction Stage)

- The product is unique; few or no competitors exist.
- Targets "Early Adopters" who value novelty over price.
- Demand is typically price inelastic because there are no close substitutes.
- Firms can set high prices (Price Skimming) to capture maximum revenue from those willing to pay a premium for the latest technology.

Pricing During Growth



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- Rivals enter the market after witnessing the pioneer's success.
- Substitutability increases, making demand more elastic.
- Firms often move from revenue-maximization to sales/market-share maximization.
- Firms may cut prices to win a dominant position and "hearts and minds" early on. This creates a strong commercial footing before the market becomes overcrowded.

Pricing During Maturity

- Ferocious competition. Sales peak, and the market is saturated with the highest number of competitors.
- Demand is at its most elastic due to the abundance of substitutes.
- Firms have little control over pricing. Competitive pressure often forces prices down to the lowest possible level.
- This stage explains why firms fight for market share during "Growth"—those without a solid base are often crushed by the "merciless price competition" of maturity.

Pricing During Decline

- Consumers lose interest and move to newer technologies. Weaker firms exit the market to seek better opportunities.
- As competitors leave, the number of substitutes falls. Demand becomes more inelastic again for the remaining "loyal" consumers.
- With less competition, price stability often returns. Remaining firms may even be able to raise prices slightly for the niche group of consumers still demanding the product.

Speculation and Price Expectations

- Definition of Speculation: Making buying or selling decisions today based on predictions of future price changes.
- Buyer Behavior:
 - Expect Prices to Rise: Buy now to avoid higher future costs.
 - Expect Prices to Fall: Wait to buy until the price drops (e.g., waiting for January sales).
- Seller Behavior:
 - Expect Prices to Rise: Wait to sell to maximize profit.
 - Expect Prices to Fall: Sell now to avoid losing value (e.g., selling a house in a declining market).

Speculation and Price Expectations

- The "Self-Fulfilling": Speculation often causes the very price change that was predicted.
 - Example: If everyone believes BP shares will rise, they buy them today. This surge in demand shifts the demand curve to the right, forcing the price to rise exactly as anticipated.
- Common Markets: Speculation is a primary driver in the Stock Exchange, Foreign Exchange (Forex), and the Housing Market.

The Working of Competitive Markets-Dealing with Uncertainty

Risk and Uncertainty in the Market

- Buyers and sellers constantly try to anticipate price changes, but they rarely operate with perfect information.
- **Risk vs. Uncertainty:**
 - **Risk:** Occurs when the **probability** of each outcome is known.
 - *Example:* A coin toss has a 50% probability of heads or tails. You can measure the variability (e.g., -£1 to +£1).
 - **Uncertainty:** Occurs when the probabilities of outcomes are **unknown** or only roughly guessed.
 - *Example:* Investing in the Stock Exchange. You might expect a share price to rise, but you cannot assign an exact mathematical probability (e.g., is it an 80% or 90% chance?).

Risk and Uncertainty in the Market

Strategies to Manage Risk

- **Stock Holding (Storage):**

- Suppliers can hold inventory to avoid being forced to sell at low market prices.
- *Example:* A wheat farmer can store grain when the harvest price is low and wait to sell until the market price rises.

- **Futures and Forward Markets:**

- Markets where buyers and sellers agree on a price today for a delivery at a specified future date.
- **For Sellers (Farmers):** Eliminates the risk of prices falling. If you lock in a price of £60 per tonne for April, you get that price even if the "spot price" (current market price) drops to £30.
- **For Buyers (Millers):** Eliminates the risk of prices rising. It allows for better budget planning by guaranteeing the purchase price months in advance.

The Economic Goals of Advertising

Advertising does more than just inform consumers about a product's existence; it is a tool to influence demand and price sensitivity.

- Objective 1: Shifting the Demand Curve to the Right:
 - By increasing desire and awareness, firms aim to sell a larger quantity at the same price.
- Objective 2: Making Demand Less Elastic (Steepening the Curve):
 - Advertisers aim to create brand loyalty, which reduces the cross-price elasticity of demand.
 - By convincing consumers that there are "no close substitutes," the firm can raise its price without suffering a significant drop in sales volume.

Questions